Answer Key - Computer Science - K-12 Level Assessment 7 (Answers are ordered exactly as in the question paper.)

# Multiple Choice

1. **Answer:** C
   *Options:* A) 4; B) 1; C) 10; D) 8
   *Note:* The correct answer is 10 because the ‘sum()‘ function in Python calculates the total of all the elements in the list, which in this case is $1 + 2 + 3 + 4 = 10$.

2. **Answer:** A
   *Options:* A) **; B) //; C) is; D) not in
   *Note:* The operator ** in Python 3 is specifically designated for performing exponential (power) calculations, raising the left operand to the power of the right operand.

3. **Answer:** B
   *Options:* A) a; B) Chemistry; C) 0; D) 1
   *Note:* The correct answer is "Chemistry" because in Python, negative indexing allows you to access elements from the end of a list, and the index -3 corresponds to the third element from the end, which is "Chemistry".

4. **Answer:** A
   *Options:* A) 1; B) 2; C) 3; D) 4
   *Note:* The correct answer is 1 because the ‘min()‘ function in Python returns the smallest value from a list, and in the list [1, 2, 3, 4], the smallest value is 1.

5. **Answer:** D
   *Options:* A) A function that averages numeric values in a column or row; B) A function that counts the values in a column or row; C) A function that rounds a numeric value; D) A function that sorts values in a column or row
   *Note:* Sorting values in a column or row allows users to easily identify outliers or discrepancies in the data set, which may indicate data entry errors.

# True / False

6. **Answer:** False
   *Options:* A) True; B) False
   *Note:* The expression evaluates to ’Chemistry’ because it accesses the element at index 1 of the list, which is the second item, not 0.

7. **Answer:** True
   *Options:* A) True; B) False
   *Note:* The statement is true because the max() function in Python 3 is specifically designed to evaluate a list of comparable items and return the one with the highest value based on their natural ordering.

8. **Answer:** True
*Options:* A) True; B) False
*Note:* The function O(log log N) grows slower than O(log N) and O(N) because as N increases, the logarithmic function grows at a much slower rate, and taking the logarithm of a logarithm results in an even slower growth rate.

9. **Answer:** True
*Options:* A) True; B) False
*Note:* The expression evaluates to 4 + (3 % 2), which simplifies to 4 + 1, resulting in the final value of 5.

10. **Answer:** False
*Options:* A) True; B) False
*Note:* The statement is false because in Python 3, the "+" operator can concatenate two strings, so "a" + "ab" results in the string "aab" without producing an error.

# Long Form Response

11. **Answer:** def sub_lists(my_list): | return subs
*Note:* The provided answer correctly identifies the need for a function that generates sublists by implying the use of a variable to hold the results, although it lacks implementation details to actually compute the sublists.

12. **Answer:** def remove_column(list 1, n): | return list 1
*Note:* correct because it defines a function named ‘remove_column‘ that takes two parameters, a nested list and a column index, but it currently lacks the implementation to actually remove the specified column from the nested list.

*End of Answer Key*